

Pruthvirajsinh Punada

iOS Developer · B.Sc. in Data Analytics

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SUMMARY

Apple-platform software engineer (Apple Developer Academy, Naples) and author of 10+ shipped projects across iOS, iPadOS, macOS, and visionOS in Swift and SwiftUI. Experienced in on-device machine learning (Core ML, Vision, SoundAnalysis) and spatial computing (RealityKit, ARKit). Shipped a published App Store app and a production website for a paying client, and placed 3rd at the Italian Ministry of Culture's data-for-cultural-heritage hackathon. Bachelor's degree in Data Analytics.

EDUCATION

Apple Developer Academy - University of Naples Federico II

Naples, Italy | 2025 - 2026

iOS Software Engineering and App Foundation Program.

University of Campania "Luigi Vanvitelli" - Bachelor's Degree, Data Analytics

2022 - 2026

Thesis: Support Vector Machine land-cover classification of 27,000 EuroSAT satellite images into 10 classes; HOG feature extraction and comparison of 3 kernels.

SKILLS

Programming Languages: Swift, Python, TypeScript, JavaScript, C++, SQL

iOS / Apple Frameworks: SwiftUI, UIKit, AppKit, RealityKit, ARKit, Core ML, Vision, SoundAnalysis, Core Haptics, Combine, SpriteKit, SceneKit, AVFoundation, Xcode

Platforms: iOS, iPadOS, macOS, visionOS

Machine Learning & Data: PyTorch, TensorFlow, scikit-learn, Computer Vision, Anomaly Detection, Data Analysis

Web & 3D: React, Next.js, Three.js, React Three Fiber, WebGL, Blender, USDZ, glTF

Tools & Practices: Git, Firebase, Vercel, REST APIs, CI/CD, MVVM, Agile

EXPERIENCE

Freelance Full-Stack & iOS Developer - Self-employed

Web & Mobile | 2025 - Present

- End-to-end design and delivery of web and native applications for paying clients and the App Store.
- Published InShorts Italia, a native iOS news aggregator with 400+ App Store downloads; aggregates 70+ Italian sources across 9 categories via a Firebase Cloud Functions backend (RSS pulled every 2 hours, plus AI summarization).
- Built and released applications across iOS, iPadOS, macOS, and visionOS using Swift and SwiftUI, owning each product from back end to UI.

Analyst & Web Developer - Pizz'Art S.r.l.s.

Caserta, Italy | 2025 - 2026

- Designed and built the company website from scratch (production Next.js, ~98% TypeScript) and managed its ongoing maintenance.
- Built the online reservation system handling customer bookings (~300/month), wired to an automated email-notification pipeline.
- Handled deployment, SEO, and ongoing updates; internationalized and deployed on Vercel behind Cloudflare to keep the site fast and reliable.

ACHIEVEMENTS

3rd Place - Cultural Heritage Data Hackathon, Italian Ministry of Culture

Istituto Centrale per la Digitalizzazione del Patrimonio Culturale (Digital Library), Rome | December 2025

- Placed 3rd of the participating teams (team of 3) at the national hackathon on AI data products for the DPaaS cultural-heritage platform, funded under EU NextGenerationEU.
- Built an AI cataloguing tool for Rome's museum manuscript collections: a curator uploads a manuscript image and a vision model auto-fills the descriptive record (estimated dating, tags, categories, and metadata), cutting manual cataloguing time for cultural operators.

PROJECTS

Pulse - On-Device Sound-to-Haptic Accessibility App (iOS)

Technologies: Swift, SwiftUI, SoundAnalysis, Core Haptics, Speech, Live Activities

- Built an accessibility app that recognizes environmental sounds (smoke alarm, doorbell, baby crying, a spoken name) entirely on-device and converts each into a distinct Core Haptics vibration and full-screen visual alert for Deaf and hard-of-hearing users.
- Engineered a fully offline pipeline (SoundAnalysis classifier + on-device Speech) with a pure, testable detection core covered by 55 automated tests and CI; no audio or data ever leaves the device.

LiveChess - Spatial Chess for Apple Vision Pro

Technologies: Swift, SwiftUI, visionOS, RealityKit, C++

- Integrated the Stockfish C++ engine into Swift for an adjustable-strength AI, move analysis, and opening book.
- Built online multiplayer over the Lichess API (OAuth PKCE authentication, live NDJSON game and event streaming, matchmaking).
- Designed the immersive 3D environments in Blender and shipped them through a USDZ-to-RealityKit pipeline; built game review and puzzle modes.

ShiftBook - Restaurant Shift & Staff Management App

Technologies: Swift, SwiftUI, Firebase, Cloud Functions

- Built a native iOS app for restaurants to schedule shifts and manage staff, with a Firebase backend (authentication and Firestore) keeping schedules in sync across the team in real time.
- Implemented role-based scheduling, staff availability, and automated shift notifications via Firebase Cloud Functions.

PinballVision - visionOS Pinball Game for Apple Vision Pro

Technologies: Swift, SwiftUI, visionOS, RealityKit, ARKit, AVAudioEngine

- Built a fully playable pinball game in a volumetric window, controlled by pinching either hand via ARKit hand tracking; the table (bumpers, slingshots, lanes) is constructed entirely from RealityKit physics bodies.
- Synthesized every sound effect in real time with AVAudioEngine (zero audio assets); added Game Center leaderboards and full VoiceOver accessibility.

Pizz'Art Caserta - Restaurant Website (client project)

Technologies: Next.js, TypeScript, React, Vercel, Cloudflare

- Designed and shipped a production Next.js website (~98% TypeScript) live for a paying restaurant client, with an online reservation flow handling ~300 reservations/month wired to an email pipeline.
- Internationalized and SEO-optimized, deployed on Vercel behind Cloudflare for performance and reliability.

SignBridge - On-Device ASL Recognition Tutor

Technologies: Swift, Core ML, Vision, SpriteKit

- Implemented real-time recognition of the full 26-letter ASL alphabet at 60 FPS, using Vision hand-pose detection feeding a custom Core ML model with prediction smoothing.
- Built spell-to-play SpriteKit mini-games; runs fully on-device with no server dependency.

BetterFinder - macOS Finder Alternative

Technologies: Swift, SwiftUI, AppKit

- Built a power-user macOS file manager bridging SwiftUI and AppKit, featuring a built-in terminal, fast file search, and syntax-highlighted preview.
- Engineered its Smart/Batch Rename engine: rule-based renaming with live preview and conflict detection, backed by 14 unit tests.

LANGUAGES

Hindi (Native), Gujarati (Native), English (Professional), Italian (Elementary)